Other Ceramic Materials

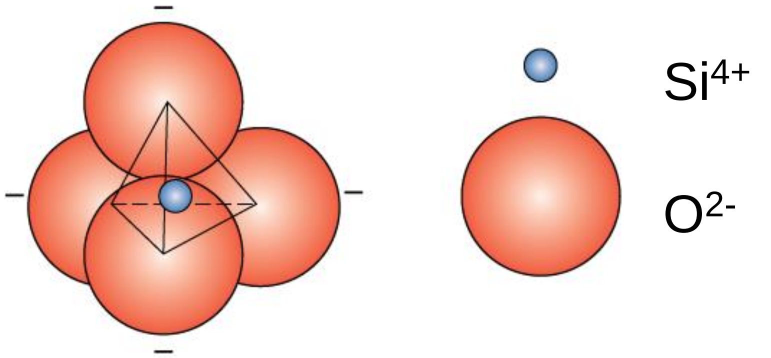
Presented by Dr Adijat ‘Wumi Inyang

Other ceramic materials

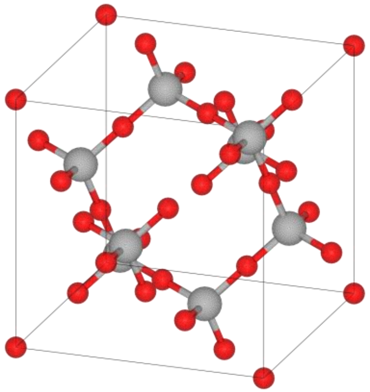
- Silica ceramics – forms of SiO_2
- Glass
- Clay mineral materials
- Cement
- Carbon materials

Silicate ceramics

Most common elements on earth are Si & O



Adapted from Figs. 4.10-11, *Callister & Rethwisch 9e*.



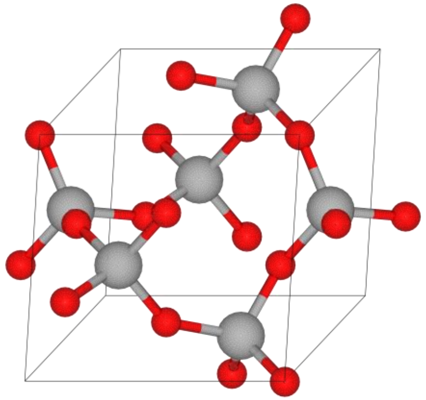
High temperature crystobalite phase

Tetragonal crystal system

Si and O form the **silica tetrahedron**
This co-ordination forms into crystalline forms

Quartz is the form found in natural rocks
It is piezo-electric with a characteristic frequency and is used in devices to measure time and mass

Crystobalite is the high temperature phase

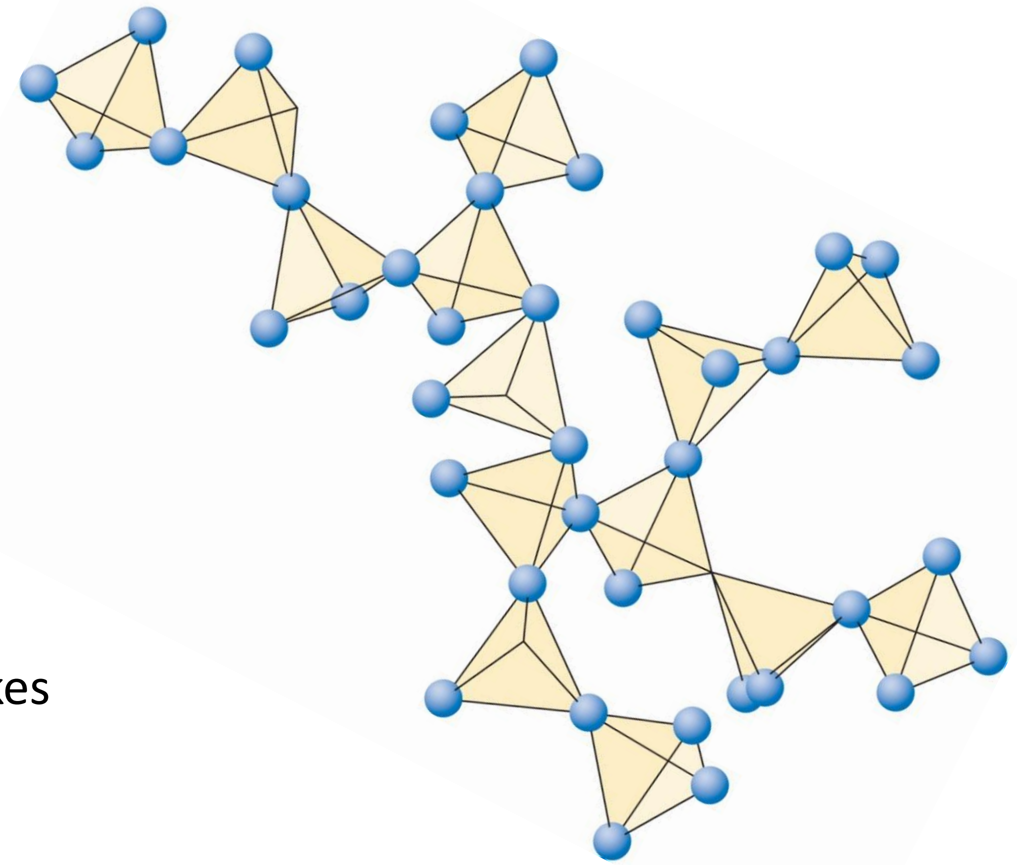


Natural quartz phase

Trigonal crystal system

Silica glasses

The **silica tetrahedra** form an amorphous 3D network of O – Si – O chains



This amorphous network of silica
is called fused quartz or **fused silica**
and is a glassy material, not crystalline

Density = 2200 kg/m^3
Softening temperature = **1665°C**
Melting temperature = **1715°C**
Coefficient of thermal expansion = **5.5 x10^{-7} / K**

This combination of low thermal expansion makes
fused silica resistant to thermal shock
Used as **high temperature window material**

Silica glasses

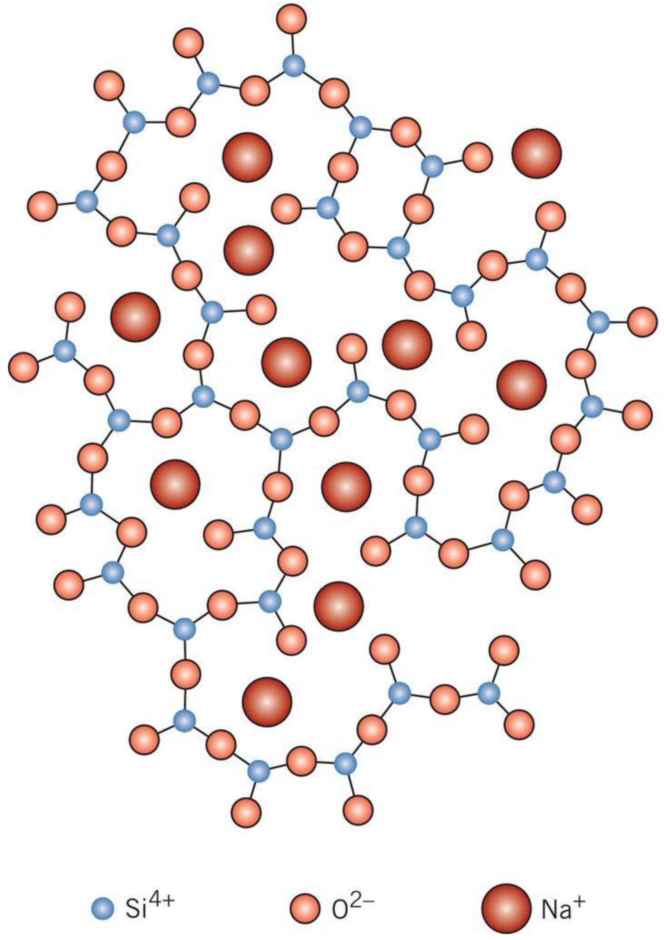
The network of O – Si – O chains are disrupted by other elements (**network modifiers**) such Na^{1+}, Ca^{2+} Al^{3+}

- The network modifiers interrupt the silica chains
- The melting temperatures are reduced

  e.g. window glass (soda-lime glass)

  Softening temperature ~ **700°C**

  Melting temperature ~ **1000°C**
- Combinations of different metal oxides are used to make a range glasses with different properties
- See Callister & Rethwisch 9e **Table 14.3**



72%SiO_2 + 14%Na_2O + 7.9%CaO +1.8%Al_2O_3 (**Soda-lime glass**)

Fig. 4.12, *Callister & Rethwisch 9e*.

Silica based glass materials

- Borosilicate glass has a low coefficient of thermal expansion **= 3 x10^{-6} / K**

  melting temperature 1650°C softening temperature 860°C

  This makes it a better choice for high temperatures but more formable than fused silica – oven wares, laboratory glassware

  80%SiO_2 + 4%NaO + 13%B_2O_3 +3%Al_2O_3 (**Borosilicate glass**)

- Some glass systems can be made to **crystallize**

  by using the correct cooling

  rate curve = **Glass-ceramics**

43.5%SiO_2 + 14%NaO + 5.5%B_2O_3
+30%Al_2O_3 + 6.5%TiO_2
(**Glass-ceramic**)

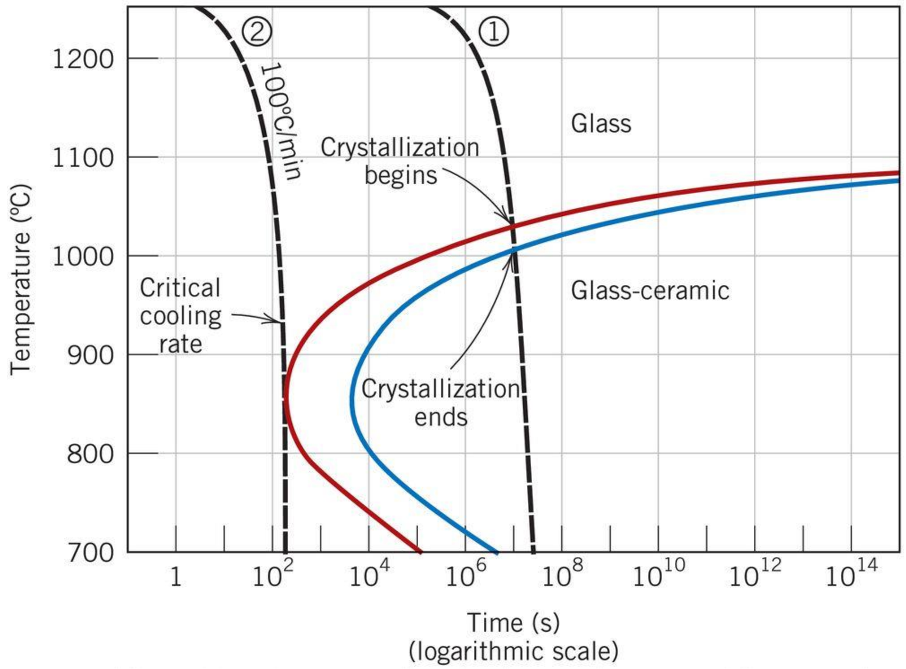
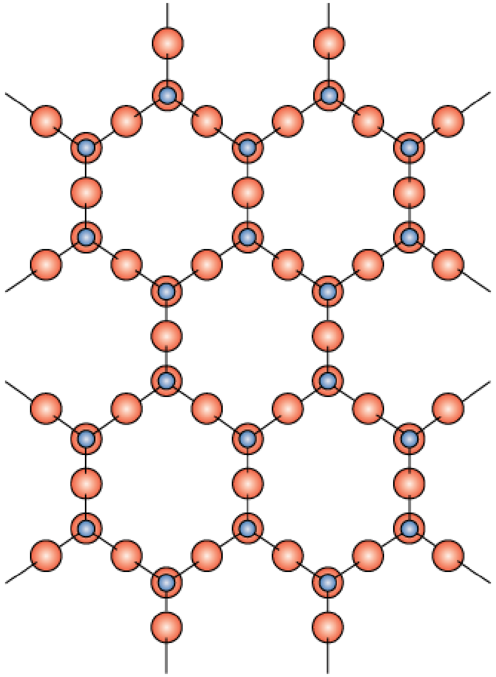


Fig. 14.15, *Callister & Rethwisch 9e*.

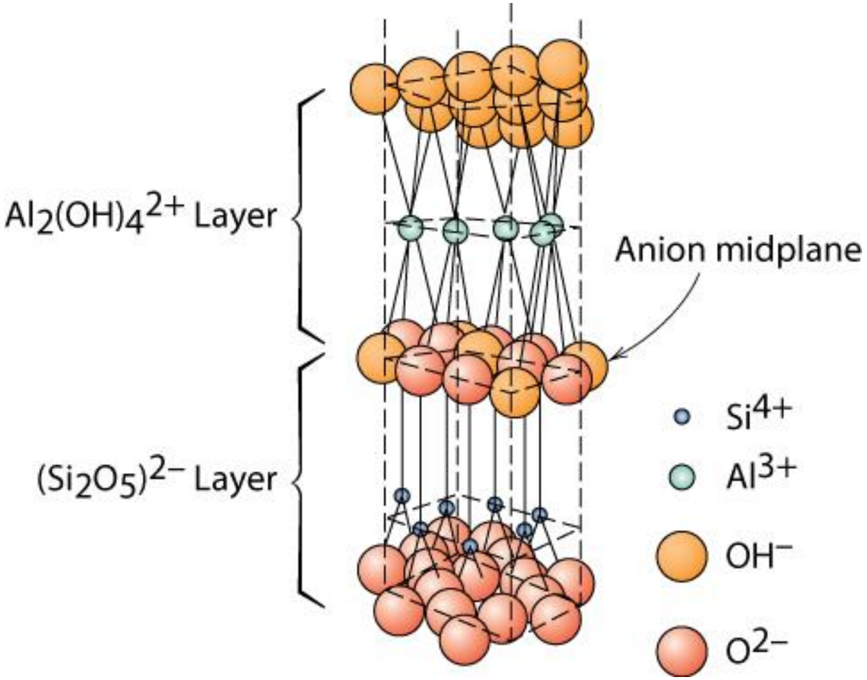
Layered Silicates

- Layered silicates (clay minerals)
  Silica tetrahedra connected together to form 2-D plane



- $(Si_2O_5)^{2-}$
- So need cations to balance charge

- Kaolinite clay alternates $(Si_2O_5)^{2-}$ layer with $Al_2(OH)_4^{2+}$ layer



Note: these sheets loosely bound by van der Waal's forces

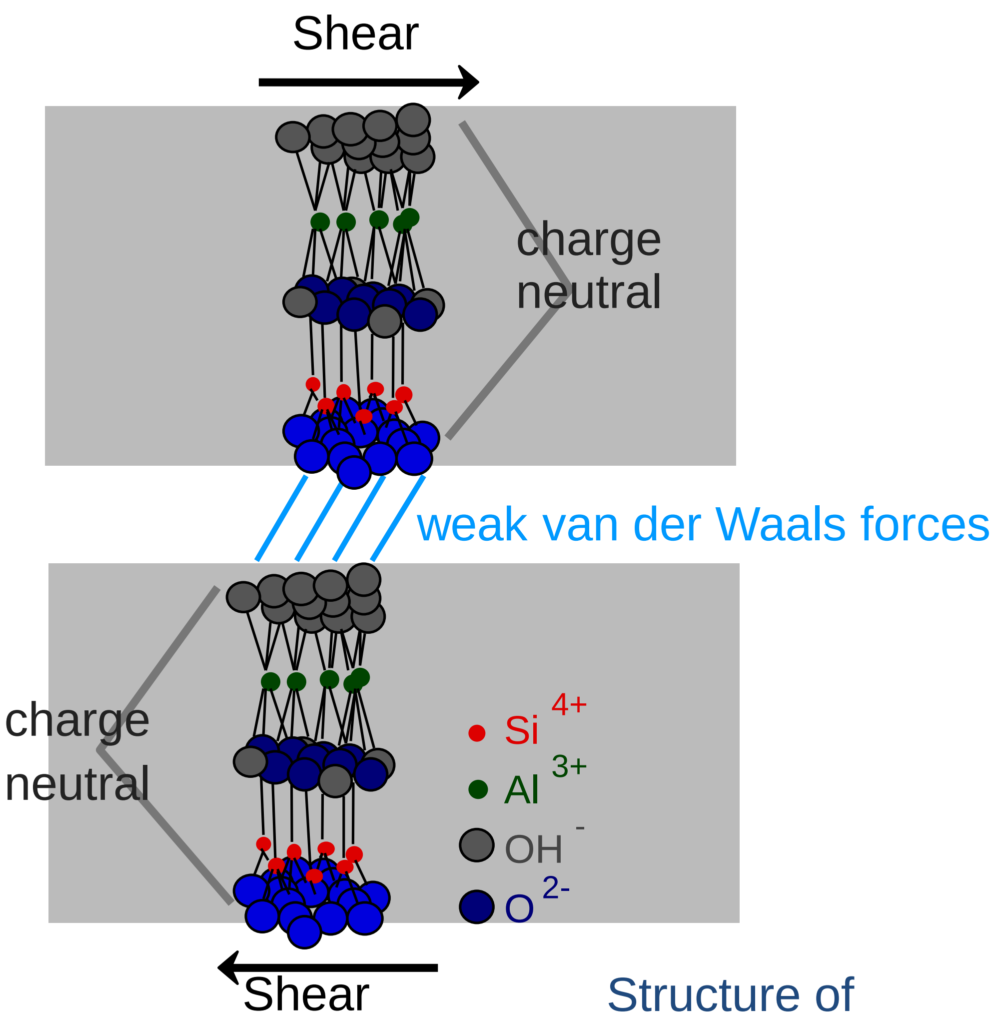
Figs. 4.14-15, *Callister & Rethwisch 9e*.

Layered Silicates

Hydroplasticity of Clay

Mechanism:

- Water molecules fit in between layered sheets
- Reduction of van der Waals bonding
- Under external force clay particles free to move past one another – becomes **hydroplastic**
- = easy to shape



Structure of Kaolinite Clay

Layered Silicates

- Can change the counter ions
  - this changes layer spacing
  - the layers also allow absorption of water
- Can exchange water for polymer molecules

  = **intercalated polymer-clay composites**

- **Micas** $KAl_3Si_3O_{10}(OH)_2$

- **Bentonite**
  - used to seal wells
  - packaged dry
  - swells 2-3 fold in H_2O
  - pump in to seal up well so no polluted ground water seeps in to contaminate the water supply

Cement

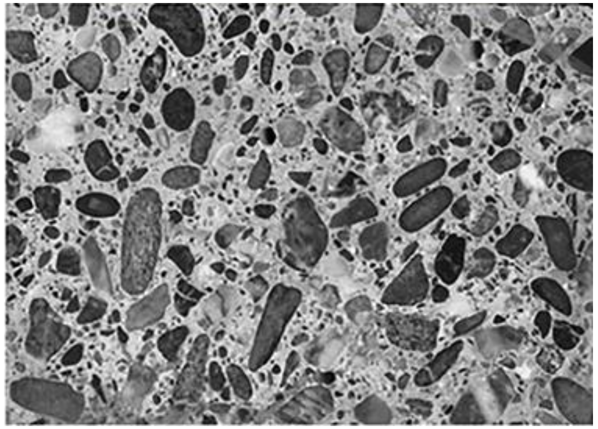
- Cement for **construction** is the most widely used product in the world at over **4000 million tonnes / year**
- High energy required to produce cement ~ 3.4 GJ/tonne
- 1 tonne of cement generates 1 tonne of CO_2

- Cement is the binder = adhesive or **matrix material in concrete**
- Known as '**hydraulic cement**' dehydrated Ca, Si, Al oxides react with water to form interlocking mineral phases and '**sets**' to become hard
- **Portland cement** is made up of **four main compounds**:
  - tricalcium silicate ($3CaO \cdot SiO_2$), [C_3S] 25-50%
  - dicalcium silicate ($2CaO \cdot SiO_2$), [C_2S] 20-45%
  - tricalcium aluminate ($3CaO \cdot Al_2O_3$) [C_3A] 5-12%
  - tetra-calcium aluminoferrite ($4CaO \cdot Al_2O_3Fe_2O_3$) [$C_4AF$] 6-12%
- The ratio of the these phases changes the rate of reaction with water and the heat evolved on setting
- The water : cement ratio (~0.3-0.4) is critical to the final strength

Concrete

- Concrete is a mixture of stone (aggregates) and sand particles held together by cement
- Concrete is a particle composite material
- A typical formulation might be

  1:2:4 cement:sand:aggregate



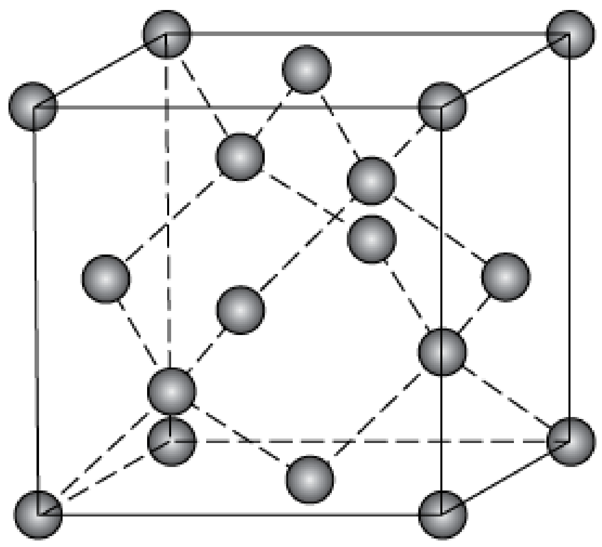
- Concrete is strong in compression ~ 40MPa but weak in tension.
- Usually reinforced with steel to resist tensile or bending forces
- Pre-stressed concrete by placing reinforcement in tension

- Easy to shape and form when first mixed with water
- **Strength develops over time**:

  50% final strength in 7 days, 100% in 28 days

Carbon Materials

- **Diamond**
  - crystalline carbon
    - hard – no easy slip planes
    - Hardest material known
    - High elastic modulus ~1150GPa
  - small diamonds
    - used for cutting tools and polishing
- **Diamond-like carbon**
  - Mixture of *sp3* and *sp2* bonds
  - Amorphous structure
  - Diamond thin films
    - hard surface coat – tools, medical devices, etc.



Diamond cubic crystal system

Fig. 4.17, *Callister & Rethwisch 9e*.

Zinc blende structure with all C atoms *sp3* hybridization produces very strong covalent character bonds a

Carbon Materials

- **Graphite**
  - layer structure – aromatic layers
  - *sp2* hybridization in plane

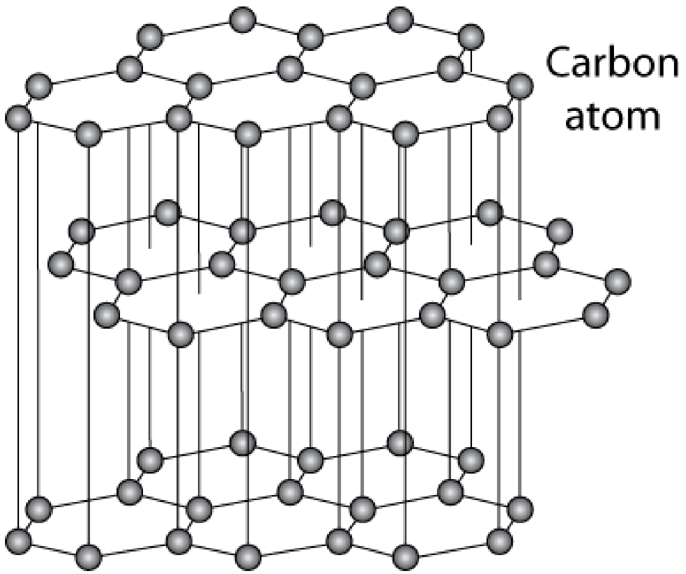
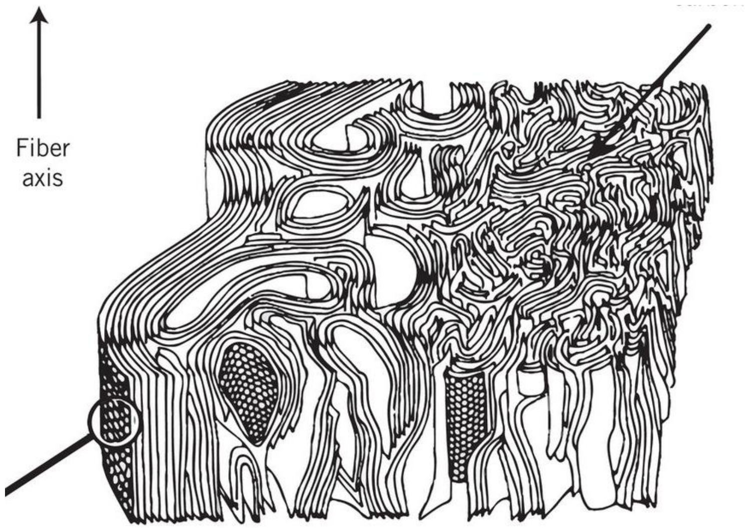


Fig. 4.18, *Callister & Rethwisch 9e*.

weak van der Waal's forces between layers
- planes slide easily, good lubricant

Graphite solid

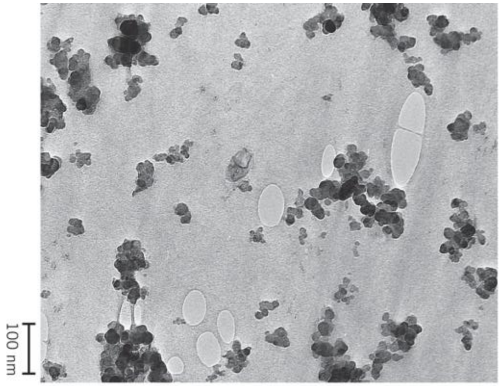
- Folded layer structure



Adapted from Fig. 14.18, *Callister & Rethwisch 9e*.

Carbon Materials

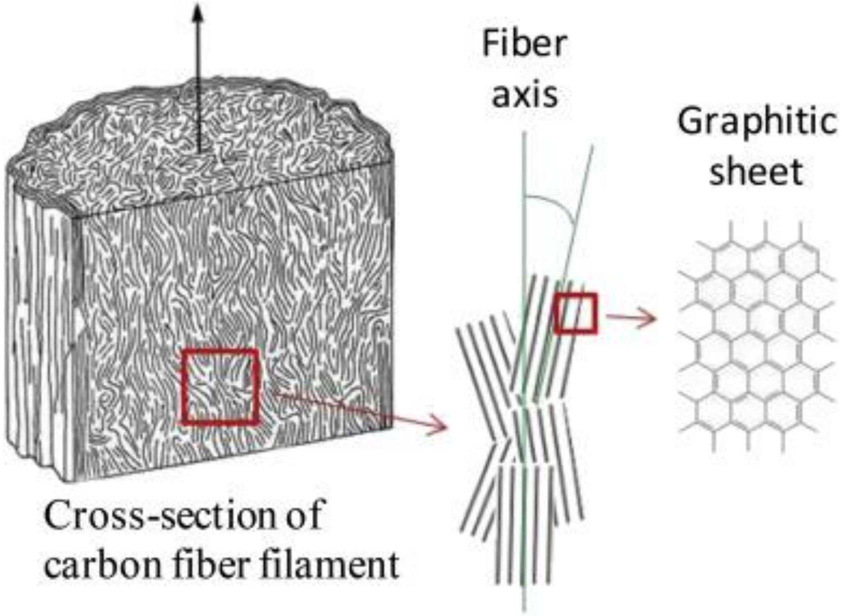
- **Carbon black**
  - Carbon nano-particles (soot)
  - High surface area > 1000m^2/g



Courtesy of Goodyear Tire & Rubber Company

Fig. 4.18, *Callister & Rethwisch 9e*.

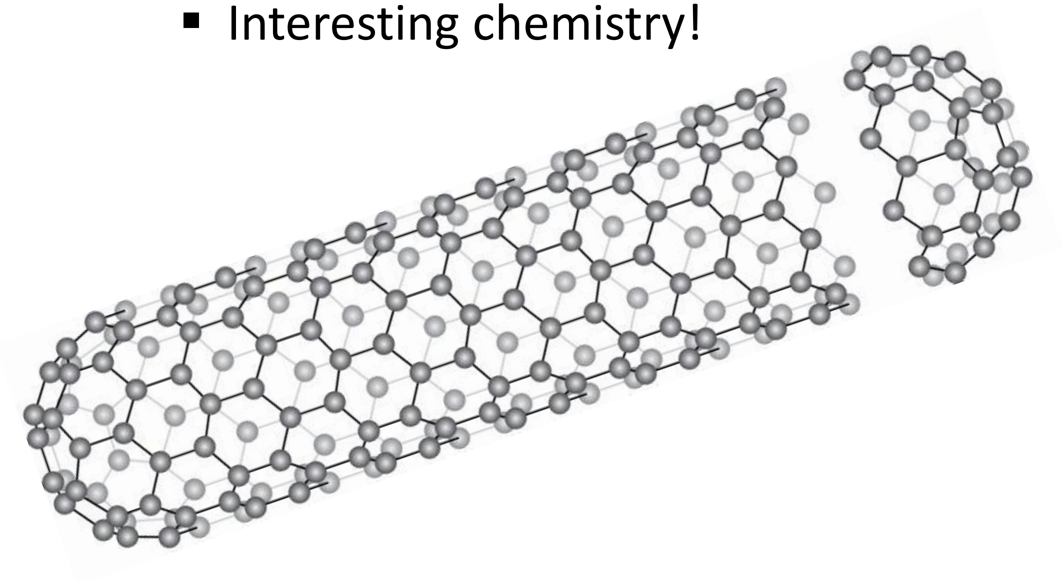
- **Carbon fibres**
  - Made from polymer fibres
  - PAN – oxidize – pyrolysis
  - Graphitic structure
  - Highly oriented at surface
  - Elastic modulus > 200GPa

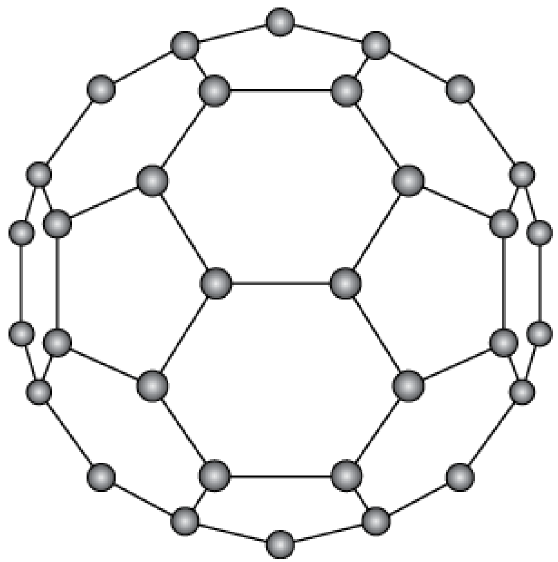


From Behr *et al*. Carbon 107, 2016 p525

Carbon Materials

- **Fullerenes** or carbon nanotubes
  - wrap the graphite sheet by curving into ball or tube
  - Buckminister fullerenes
    - Like a soccer ball C_{60} - also C_{70} + others
    - Can trap atoms or molecules inside
    - Interesting chemistry!



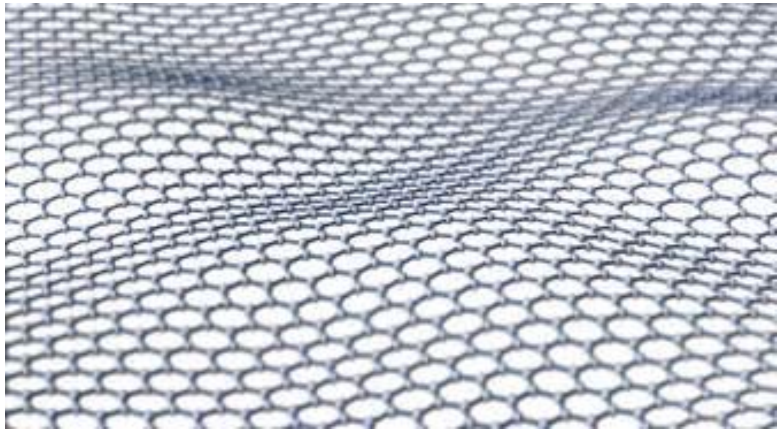


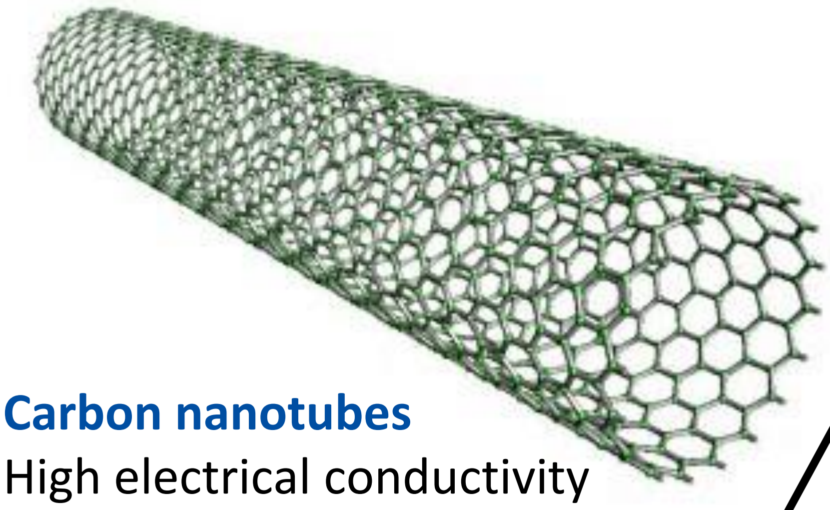
Adapted from Figs. 12.18 & 12.19, *Callister 7e.*

Carbon Materials

Carbon materials are becoming very important in new technologies
Quantum states that give unusual properties

Graphene
Single layer graphene
Very high stiffness in the layer





Carbon nanotubes
High electrical conductivity along the tube

1.2nm diameter
Single walled nanotube
(Most are multi-walled)

Summary

Other ceramic materials
(or in-organic, non-metallic materials):

- **Silica based materials**
  - Crystalline phases
  - Glasses
  - Layered solids – clay minerals

- **Cement**
  - Hydraulic cements from metal oxides
  - Basis of concrete (particle composite)

- **Carbon based materials**
  - Diamond and diamond-like materials
  - Graphite and graphene materials
  - Fullerenes and carbon nanotubes